Vulnerability Assessment and Risk Analysis

Course: Google Cybersecurity Professional Certificate
Project Type: Vulnerability Assessment
Framework: NIST SP 800-30 Rev. 1
Environment: Simulated E-Commerce Organization

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1. System Description

The system being assessed is a database server used by an e-commerce company.

The server has a powerful CPU and 128 GB of memory. It runs Linux and hosts a MySQL database management system.

The server uses IPv4 networking, communicates with other servers, and supports SSL/TLS encrypted connections.

2. Scope

The assessment focuses on the database server's access controls and potential security risks over a three-month period, from June to August of the simulated assessment year.

NIST SP 800-30 Rev. 1 was used to guide the qualitative risk assessment.

3. Purpose

The database server is an important business asset because it stores information needed for company operations and supports employees accessing data remotely.

Protecting the server is necessary to prevent unauthorized access and maintain business continuity.

If the server becomes unavailable or compromised, business activities that depend on its data and network connections could be disrupted.

4. Risk Assessment

The following table documents the threat sources, events, and risk scores from my original assessment.

Risk was calculated using:

Risk = Likelihood × Severity

Both likelihood and severity were scored from 1 to 3.

Threat Source	Threat Event	Likelihood	Severity	Risk
Hacker	Credential theft resulting in unauthorized access, modification, or deletion of critical information	3	3	**9**
CPU / Hardware Failure	Processor becomes outdated, is not maintained, or stops functioning	2	2	**4**

Reference Example From the Original Template

The course template also included the following example, which was not an additional threat I independently identified.

Threat Source	Threat Event	Likelihood	Severity	Risk
Competitor	Exfiltration of sensitive information	3	3	9

5. Assessment Approach

My assessment focused primarily on external threats because the database server had been publicly accessible for an extended period.

This exposure increased the possibility of malicious actors attempting to access sensitive information or compromise the server.

I also considered hardware failure because the server's processor is important for supporting business operations.

Although the processor is powerful, hardware failures could interrupt services and affect organizational productivity.

6. Remediation Strategy

The main recommendation was to restrict public access to the database server.

Access should be limited to authorized employees and stakeholders who require it for business operations.

This would reduce unnecessary exposure to external threats and lower the likelihood of unauthorized access.

I also recommended maintaining the server's hardware and conducting routine checks to reduce the risk of system failure and operational disruption.

7. Skills Demonstrated

- Vulnerability Assessment
- Qualitative Risk Analysis
- NIST SP 800-30 Rev. 1
- Threat Identification
- Risk Scoring
- Access Control
- Security Remediation Planning
- Business Continuity Considerations

8. Conclusion

This project provided experience identifying security risks affecting a publicly accessible database server.

I evaluated potential external threats and hardware-related risks, calculated risk scores, and proposed security measures to improve access control and system reliability.

The assessment highlighted the importance of restricting unnecessary public access and maintaining critical infrastructure.

This project was completed as part of the Google Cybersecurity Professional Certificate using a simulated e-commerce scenario. The risk scores and recommendations reflect my original coursework, with wording and formatting refined for portfolio presentation.